



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE A

Course Code & Title: CS3591 Computer Networks

Name of the Faculty member (s): Mrs.M.Swarna Sudha

Innovative Practice Description

Unit / Topic: Unit V / Network Performance

Course Outcome: CO5

Topic Learning Outcome: 5e

Activity Chosen: Virtual Laboratory

Justification:

The Virtual Laboratory is an interactive simulation playground where users can design and carry out simulated experiments. An on-screen simulator that allows students to test theories and see outcomes. They give the student the opportunity to work with lab equipment that functions essentially the same as it would in an actual setting. This involves creating and conducting a series of experiments within the virtual laboratory to assess key performance metrics such as latency, bandwidth, and packet loss. The goal is to derive actionable insights that can inform optimization strategies for the actual network. This virtual lab also offers benefits such as reduced software and hardware costs for setting up an actual advanced network technology lab, improved learning by interactive visualization

- **Time Allotted for the Activity:** 20 minutes

Details of the Implementation:

- Instructor explained the particular concepts in classroom within 35 minutes..
- The following link is used to access the virtual lab.
<http://vlabs.iitkgp.ernet.in/ant/3/>
- The instructor demonstrated specific operations within the Virtual Lab using simulation buttons. This hands-on demonstration likely included showing students how to perform certain tasks, configure network settings, or troubleshoot issues using the simulator's features.
- For example, the use of simulation buttons could involve configuring routers, adjusting network parameters, initiating network protocols, or analyzing network performance metrics. The demonstration serves as a practical illustration of the theoretical concepts discussed in the classroom.

- The instructor highlighted key features of the simulator, such as the ability to simulate different network conditions, visualize data flow
- The figure 1 shows the integration of theory, simulation, and self-evaluation.

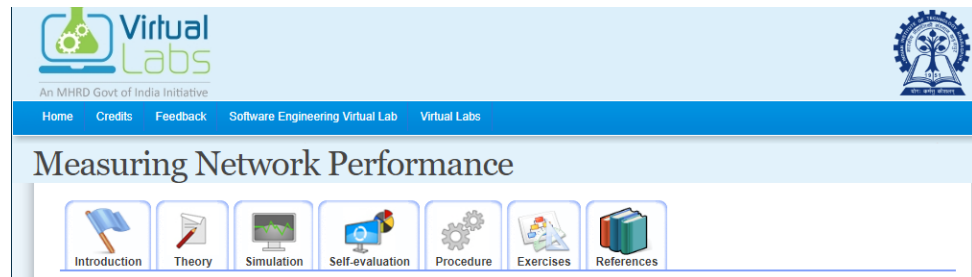


Figure 1 shows the integration of theory, simulation, and self-evaluation.

- Step-by-step instructional model implemented in a virtual laboratory setting was given in detail in the figure 2. The model involves a systematic approach where students receive detailed instructions for theoretical concepts, engage in hands-on simulations within a virtual environment, and follow step-by-step guidance

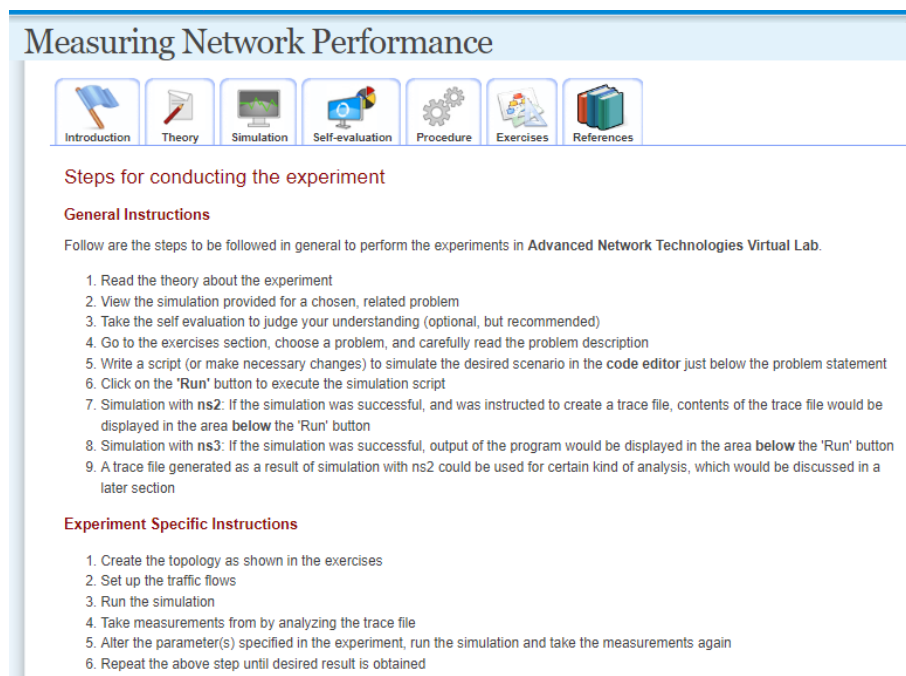


Figure 2 Step by step instruction for simulation

Measuring Network Performance



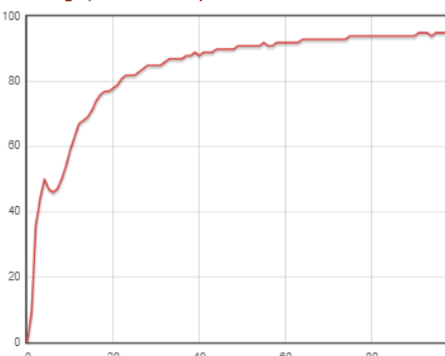
Consider a local area network connecting node numbers 3, 4, 5. This LAN is connected to two external servers (node no.0 and node no.1) through a router (node no.2).

Here in this network node no. 0 is connected to node no.2 by a duplex link and node no.1 is connected to node no.2 by a duplex link. Again, node no.2 and node no.3 are connected to each other with two simplex links thereby allowing bi-directional data transmission.

The server at node no.0 is running an application on TCP/IP and sending data to node no.4.

For the two simplex links between node 2 and node 3, different possible values for the link bandwidth, propagation delay, queue type are shown in the right hand side panel.

Throughput vs time plot



Network Parameters	
TCP Type	Newreno
Packet Size	1000 bytes
Bandwidth	☒ 0.3 Mb ☐ 0.6 Mb ☐ 0.9 Mb
Propagation Delay	☒ 25 ms ☐ 50 ms ☐ 100 ms
Queue Type	☒ DropTail ☐ RED

Figure 3 performance of the network with different parameters

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PSO1
CO 5	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Students can able to apply Logical thinking and make the student able to think about complex engineering problems.	Students will be able to approach the problem by applying the they learnt and solve the flexible changes to the network setup, enabling iterative testing and quick adaptation to evolving requirements.	Students will be able to interpret the complex engineering problems using valuable insights into network behavior	The problem solving skill earned through this activity helps the students in solving various problems in information and communication engineering.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

This helped the students to understand the

- The concept of network performance evaluation, and different related metrics
- Overview on bandwidth sharing by multiple traffic flows
- Identify bottlenecks in a network

- ❖ ***Benefit of the practice:***

- Based on the activity student's felt that it was easy to understand the concept.
- Most of the students were actively participated and enjoyed once they answered correctly in self-assessment (http://vlabs.iitkgp.ernet.in/ant/3/self_evaluation/)

- ❖ ***Challenges faced in implementation:***

- Some students only involved in the demonstration activity.
- Some students faced initial challenges in navigating the virtual laboratory interface. Additional support and training were provided to address this concern.
- Balancing theory, simulations, and self-evaluation requires effective time management, an aspect that was emphasized in subsequent sessions.
- The same activity will be planned in the laboratory session to make all the students familiar in the virtual environment.

References:

1. <https://python-iitk.vlabs.ac.in/exp/arithmetic-operations/index.html>
2. <https://www.ritrjpm.ac.in/images/computer-science/Virtual%20Lab-Routing%20in%20MANET.pdf>
3. <https://www.techtarget.com/searchnetworking/definition/network-performance-monitoring>